

Bhanuday Sharma

PhD (Aerospace) – IIT Kanpur

bhanuday@iitk.ac.in ✉

+91-9509988506 ☎

bhanuday.sharma@gmail.com ✉

www.bhanuday.com 🌐

EDUCATIONAL QUALIFICATIONS			
Year	Degree	Board/Institution	Percentage/CPI
2016-present	MTech+PhD (Aerospace)	Indian Institute of Technology Kanpur, Kanpur	9.2
2016	BTech (Aerospace)	University College of Engineering, RTU, Kota	72.41% (Honors)
2011	Twelfth Class	Board of Secondary Education, Rajasthan	84.31%
2008	Tenth Class	Board of Secondary Education, Rajasthan	88.83%
SCHOLASTIC ACHIEVEMENTS			
• 5 journal articles (1 published, 1 submitted, 3 manuscript under-preparation) and 6 conference presentations • During the PhD, mentored three MTech thesis, two BTech projects, and 12 internships • Scored All India Rank 56 in GATE-2016 • Secured 1st position in inter-college physics competition in April 2013 • Best Design Award in workshop on Quadcoptors • Secured rank-1 in the competition of Robowars at THAR-15			
THESIS AND PROJECTS			
PhD Thesis	Estimation of bulk viscosity of dilute gases Supervisor: Dr. Rakesh Kumar (<i>In collaboration with DELL EMC R&D, Bengaluru, India</i>) • Estimated bulk viscosity of dilute gases using molecular dynamics simulations • Proposed and demonstrated a novel first principle approach for the same (Aug'17-present)		
Machine Learning Projects	Optimization of intermolecular interaction potential with non-linear ML regression (<i>Part of PhD thesis</i>) • The objective was to reduce the number of interaction sites between N₂ & H₂O molecules to reduce the computational cost of molecular dynamics simulations • Employed various optimization algorithms, viz., trust-region methods, genetic algorithms, stochastic gradient descent, RMSProp, ADAM, and Levenberg-Marquardt algorithm in Scikit-Learn and PyTorch frameworks • Parallelized the CPU-based code using the multiprocessing module and GPU-based code using CuPy and PyTorch (Feb'20-Feb'21)		
	Application of physics informed neural networks (PINN) to solve fluid flows (<i>Master's thesis</i>) (<i>In collaboration with NVIDIA Corporation, India</i>) • Objective is to use PINN to solve partial differential equations arising in various fluid flow problems, including Burger's equation, Navier-Stokes equation, and Boltzmann equation (May'21-present)		
	Morphological analysis of Hindi language using machine learning tools • The aim was to classify words in the categories like gender, noun, and identify its word morphemes. • A Python script was developed using k-Means clustering algorithm and SVM classification algorithm. (May'19-Jul'19)		
Other Projects	• CFD Analysis of can combustor (B. Tech Project) (Dec'15-Feb'16) • Development of unmanned aircraft system (UAS) for crop dusting purpose (Mar'17-Apr'17) • Development of optical character recognition (OCR) software in LabVIEW (Jun'17-Jul'17) • Calculation of shear viscosity using molecular dynamics simulations (Jan'16-Apr'16)		
Internship	Characterization of can combustor (<i>National Aerospace Laboratory, Bangalore, India</i>) • Performed experimental measurements for performance optimization of a turbojet combustion chamber (May'15-Aug'15)		
KEY SKILLS			
Programming Languages		Basics of C, C++, Python, HTML, Linux BASH Programming, VBA	
Python Libraries		Numpy, CuPy, Scipy, Pandas	
Machine Learning Frameworks		PyTorch, Keras, Scikit Learn, Excel	
Parallel Programming		OpenMP, OpenMPI, Cuda	
HPC and DevOps Skills		Comfortable with SLURM, PBS Pro, software compilation/building from source on Linux platform, and benchmarking	
Simulation Techniques		Computational fluid dynamics (CFD), Molecular Dynamics (MD), Lattice Boltzmann Method (LBM), Direct Simulation Monte Carlo (DSMC), Finite Element Method (FEM)	
Research Skills		Strong analytical and problem-solving skills, literature survey, technical writing, and presentation skills	
POSITIONS OF RESPONSIBILITY			
Chairman – mess committee, Hall-4, IIT Kanpur • Ensured proper management and quality maintenance of a mess serving more than 450 student • Handled the tendering process of procurement of raw materials worth Rs. 10 lacks every month			(Feb'19-Feb'20)
Student coordinator and founding member of Vivekananda Kendra, UCE, RTU, Kota • Along with two other student coordinatorestablished Vivekanand Kendra in the college, which is working for the enhancement of moral and ethical values among students. The Kendra provides free tuition to needy students of nearby economically backward colonies.			(Feb'14-Apr'15)
Coordinator –hospitality and registration committee, THAR-15 • Recruited and led a 15 member team in university-level technical festival THAR'15 • Ensured proper residence and smooth registration of more than 200 people during the festival			(Feb'14-Apr'15)